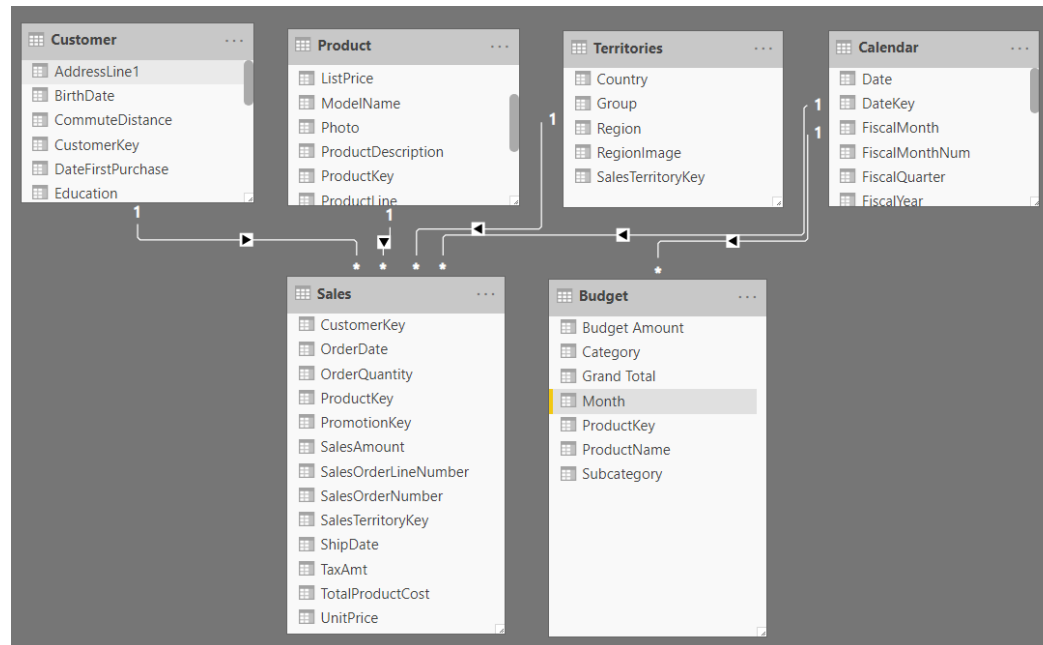


Business Intelligence Solutions using DAX

Introduction

In this session, you will use the Power BI model "AdventureWorksTask" to do Data Analytics in Power BI. **You worked on that file in the previous task.** If you do not have that file, make sure you go back to the previous task and complete it. The image below shows the model that we are going to use for Data Analytics in this workbook. The model has 6 tables:



- **Calendar.** This table shows information about each date (e.g. day, month, year, day of the week, quarter, etc.). This is going to be useful when we need to filter the transactions based on a specific period of time.
- **Customer.** This table stores the information about customers (e.g. name, surname, age, birthdate, etc.)
- **Product.** This table reports the information about the products available (e.g., the product name, category, model name, etc.)
- **Sales.** This is the most important table of the model. It reports the details of each transaction (e.g. product sold, sales amount, ship date, etc.)
- **Territories.** This table reports the details of territories where the products have been sold (e.g. region, country, etc.)
- **Budget.** This table reports the budget for each month in 2016 for different products.

In this workbook, you will learn how to add calculated columns and measures using DAX in Power BI. Before doing that, you can find below a quick recap on DAX.

A quick recap on DAX

Within Power BI Desktop there is sometimes the need to take the data you have and create new columns or measures from this data. DAX is a language within Power BI, that allows you to do just that very easily. We have already seen some example of DAX in the lectures, and below there is a recap.

DAX stands for **Data Analysis eXpressions**, it is a collection of functions, operators, and constants, used to calculate an expression or formula and return a desired value from it, whether that is a new column or a new measure.

Understanding simple functions

The syntax of a simple function follows these rules.

Output name = **function**(**table name**[**column name2**])

When following this:

Output name – is what you want to name the output (this could be a column for example) that you are wanting the information to go into.

Function – is what you are wanting to do to the data (for example AVERAGE or SUM).

Table name – is where the column we are using the data from to manipulate

Column name2 – is the column name of where the data is on the table that we want to manipulate.

An example of this is the formula below:

Total = **SUM**(**Sales**[**SalesAmount**]) : this formula creates a new expression **Total** (we call this “Measure”). The “Total” stores a number which is the SUM of the SalesAmount column in the Sales table. In this example, SUM is the function, Sales is the table name and SalesAmount is the name of the column in that table.

Different types of functions

There are different types of functions, these are:

Date and time: datediff, now, today, etc.

Time intelligence: datesbetween, firstdate, lastdate, etc.

Information: contains, customdata, lookupvalue, etc.

Logical: and, or, iferror, etc.

Mathematical: abs, acos, degrees, etc.

Statistical: average, count, max, etc.

Text: concatenate, combinevalues, etc.

Parent/child: path, pathcontains, etc.

Table: filter, values, distinct, etc.

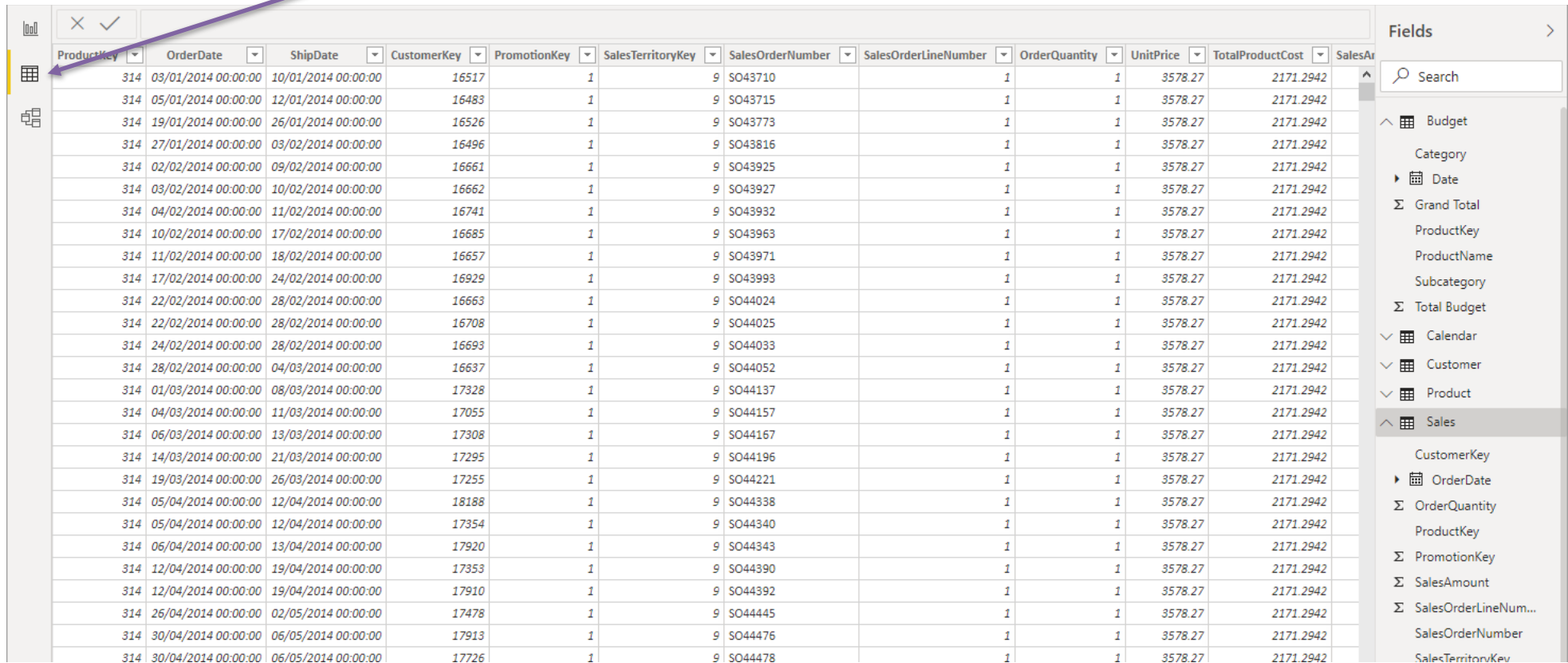
Other: datatable, groupby, error, etc.

For some of the common ones please check out <https://docs.microsoft.com/en-us/dax/> website.

We are now going to use DAX formulae to create calculated columns and measures into our Data Model.

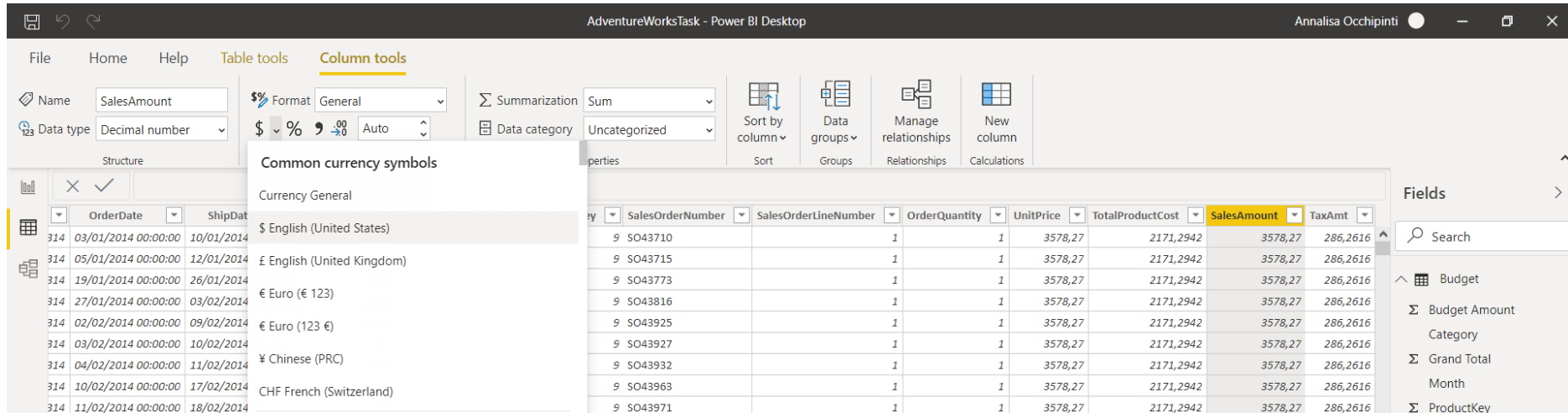
Step 1: Adding Calculated Columns

Thanks to the steps done in the previous tutorial, our data is ready to be used for Data Analytics and we will be using DAX (Data Analysis Expressions) to do so. The first step is to open the Sales table in the main Power BI window as shown below:



ProductKey	OrderDate	ShipDate	CustomerKey	PromotionKey	SalesTerritoryKey	SalesOrderNumber	SalesOrderLineNumber	OrderQuantity	UnitPrice	TotalProductCost	SalesAmount
314	03/01/2014 00:00:00	10/01/2014 00:00:00	16517	1	9	SO43710	1	1	3578.27	2171.2942	
314	05/01/2014 00:00:00	12/01/2014 00:00:00	16483	1	9	SO43715	1	1	3578.27	2171.2942	
314	19/01/2014 00:00:00	26/01/2014 00:00:00	16526	1	9	SO43773	1	1	3578.27	2171.2942	
314	27/01/2014 00:00:00	03/02/2014 00:00:00	16496	1	9	SO43816	1	1	3578.27	2171.2942	
314	02/02/2014 00:00:00	09/02/2014 00:00:00	16661	1	9	SO43925	1	1	3578.27	2171.2942	
314	03/02/2014 00:00:00	10/02/2014 00:00:00	16662	1	9	SO43927	1	1	3578.27	2171.2942	
314	04/02/2014 00:00:00	11/02/2014 00:00:00	16741	1	9	SO43932	1	1	3578.27	2171.2942	
314	10/02/2014 00:00:00	17/02/2014 00:00:00	16685	1	9	SO43963	1	1	3578.27	2171.2942	
314	11/02/2014 00:00:00	18/02/2014 00:00:00	16657	1	9	SO43971	1	1	3578.27	2171.2942	
314	17/02/2014 00:00:00	24/02/2014 00:00:00	16929	1	9	SO43993	1	1	3578.27	2171.2942	
314	22/02/2014 00:00:00	28/02/2014 00:00:00	16663	1	9	SO44024	1	1	3578.27	2171.2942	
314	22/02/2014 00:00:00	28/02/2014 00:00:00	16708	1	9	SO44025	1	1	3578.27	2171.2942	
314	24/02/2014 00:00:00	28/02/2014 00:00:00	16693	1	9	SO44033	1	1	3578.27	2171.2942	
314	28/02/2014 00:00:00	04/03/2014 00:00:00	16637	1	9	SO44052	1	1	3578.27	2171.2942	
314	01/03/2014 00:00:00	08/03/2014 00:00:00	17328	1	9	SO44137	1	1	3578.27	2171.2942	
314	04/03/2014 00:00:00	11/03/2014 00:00:00	17055	1	9	SO44157	1	1	3578.27	2171.2942	
314	06/03/2014 00:00:00	13/03/2014 00:00:00	17308	1	9	SO44167	1	1	3578.27	2171.2942	
314	14/03/2014 00:00:00	21/03/2014 00:00:00	17295	1	9	SO44196	1	1	3578.27	2171.2942	
314	19/03/2014 00:00:00	26/03/2014 00:00:00	17255	1	9	SO44221	1	1	3578.27	2171.2942	
314	05/04/2014 00:00:00	12/04/2014 00:00:00	18188	1	9	SO44338	1	1	3578.27	2171.2942	
314	05/04/2014 00:00:00	12/04/2014 00:00:00	17354	1	9	SO44340	1	1	3578.27	2171.2942	
314	06/04/2014 00:00:00	13/04/2014 00:00:00	17920	1	9	SO44343	1	1	3578.27	2171.2942	
314	12/04/2014 00:00:00	19/04/2014 00:00:00	17353	1	9	SO44390	1	1	3578.27	2171.2942	
314	12/04/2014 00:00:00	19/04/2014 00:00:00	17910	1	9	SO44392	1	1	3578.27	2171.2942	
314	26/04/2014 00:00:00	02/05/2014 00:00:00	17478	1	9	SO44445	1	1	3578.27	2171.2942	
314	30/04/2014 00:00:00	06/05/2014 00:00:00	17913	1	9	SO44476	1	1	3578.27	2171.2942	
314	30/04/2014 00:00:00	06/05/2014 00:00:00	17726	1	9	SO44478	1	1	3578.27	2171.2942	

In this table, the **SalesAmount** and **TaxAmt** (TaxAmount) are separate columns (the last two columns). These columns show different amounts of money. We want to add the "\$" symbol next to each entry. To do so, select the **SalesAmount** column and change the formatting by clicking on the Column Tools ribbon (Modelling ribbon, in the old version) and press the drop-down menu next to the dollar symbol in the Formatting section. Click on English (United States) as shown below. This will change the format of your column. Repeat the same steps for the **TaxAmt** column. Repeat the same steps for the **Budget Amount** column in the Budget table.



The screenshot shows the Power BI Desktop interface with the 'Column tools' ribbon selected. The 'Format' dropdown is open, showing 'Common currency symbols'. The 'SalesAmount' column is selected in the data table, and the 'TaxAmt' column is also visible. The 'Fields' pane on the right shows the 'Budget' table with fields like 'Budget Amount', 'Category', 'Grand Total', 'Month', and 'ProductKey'.

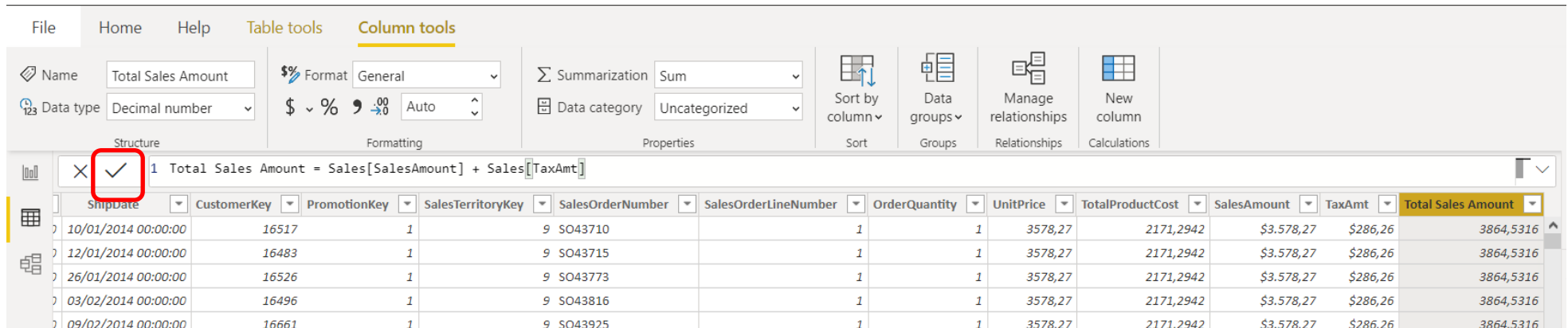
OrderDate	ShipDate	SalesOrderNumber	SalesOrderLineNumber	OrderQuantity	UnitPrice	TotalProductCost	SalesAmount	TaxAmt
03/01/2014 00:00:00	10/01/2014	9 SO43710	1	1	3578,27	2171,2942	3578,27	286,2616
05/01/2014 00:00:00	12/01/2014	9 SO43715	1	1	3578,27	2171,2942	3578,27	286,2616
19/01/2014 00:00:00	26/01/2014	9 SO43773	1	1	3578,27	2171,2942	3578,27	286,2616
27/01/2014 00:00:00	03/02/2014	9 SO43816	1	1	3578,27	2171,2942	3578,27	286,2616
02/02/2014 00:00:00	09/02/2014	9 SO43925	1	1	3578,27	2171,2942	3578,27	286,2616
03/02/2014 00:00:00	10/02/2014	9 SO43927	1	1	3578,27	2171,2942	3578,27	286,2616
04/02/2014 00:00:00	11/02/2014	9 SO43932	1	1	3578,27	2171,2942	3578,27	286,2616
10/02/2014 00:00:00	17/02/2014	9 SO43963	1	1	3578,27	2171,2942	3578,27	286,2616
11/02/2014 00:00:00	18/02/2014	9 SO43971	1	1	3578,27	2171,2942	3578,27	286,2616

Now that we have changed the format of the data, the next step is to add the SalesAmount and TaxAmt columns together to work out the total Amount (which will be stored in a new column). Click on New Column in the Column Tools (or in the Modelling ribbon, for the Power BI old versions) and in the formula bar copy the following DAX expression (as shown in the next image):

Total Sales Amount = Sales[SalesAmount] + Sales[TaxAmt]

where Total Sales Amount is the name of the new column that will be added to the table. Once you are done, click on the tick next on the left of the formula bar or press Enter in your keyboard. You should now see a table like the one below:

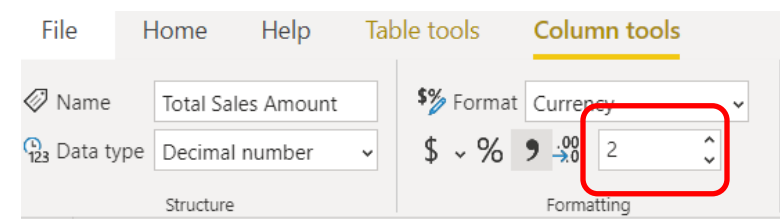
This formula performs a row-by-row sum. It sums the value from the SalesAmount column in the Sales table (expressed as Sales[SalesAmount]) and the TaxAmt column from the Sales tab (expressed as Sales[TaxAmt]). The resulting value is saved in the corresponding row of the new column, i.e. Total Sales Amount (on the left of the equal sign). The common syntax in DAX is `tableName[column_in_that_table]`



The screenshot shows the Power BI interface with the 'Column tools' ribbon active. The formula bar contains the DAX expression: `Total Sales Amount = Sales[SalesAmount] + Sales[TaxAmt]`. A red box highlights the checkmark icon on the left of the formula bar. Below the formula bar, a table is displayed with the following columns: ShipDate, CustomerKey, PromotionKey, SalesTerritoryKey, SalesOrderNumber, SalesOrderLineNumber, OrderQuantity, UnitPrice, TotalProductCost, SalesAmount, TaxAmt, and Total Sales Amount. The 'Total Sales Amount' column is highlighted in yellow. The table contains five rows of data, all showing a total sales amount of 3864,5316.

ShipDate	CustomerKey	PromotionKey	SalesTerritoryKey	SalesOrderNumber	SalesOrderLineNumber	OrderQuantity	UnitPrice	TotalProductCost	SalesAmount	TaxAmt	Total Sales Amount
10/01/2014 00:00:00	16517	1	9	SO43710	1	1	3578,27	2171,2942	\$3.578,27	\$286,26	3864,5316
12/01/2014 00:00:00	16483	1	9	SO43715	1	1	3578,27	2171,2942	\$3.578,27	\$286,26	3864,5316
26/01/2014 00:00:00	16526	1	9	SO43773	1	1	3578,27	2171,2942	\$3.578,27	\$286,26	3864,5316
03/02/2014 00:00:00	16496	1	9	SO43816	1	1	3578,27	2171,2942	\$3.578,27	\$286,26	3864,5316
09/02/2014 00:00:00	16661	1	9	SO43925	1	1	3578,27	2171,2942	\$3.578,27	\$286,26	3864,5316

If you want to change the format of your column (add the units or remove the decimal digits), select the column and click on the Column tools tab and select “\$ English (United States)” as done before (this can be found under the Modelling ribbon in the old versions of Power BI). To change the number of decimal digits displayed, simply enter the number of digits you want to keep after the decimal point in the box at the bottom right corner of the Formatting section (see right). Power BI might have already set this equal to 2 decimal digits. However, you can change this if you want. I left mine as 2 in this tutorial.

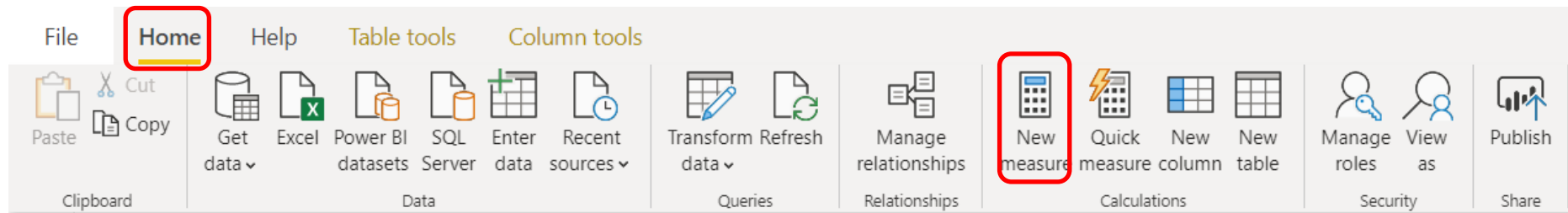


Step 2: Adding Measures

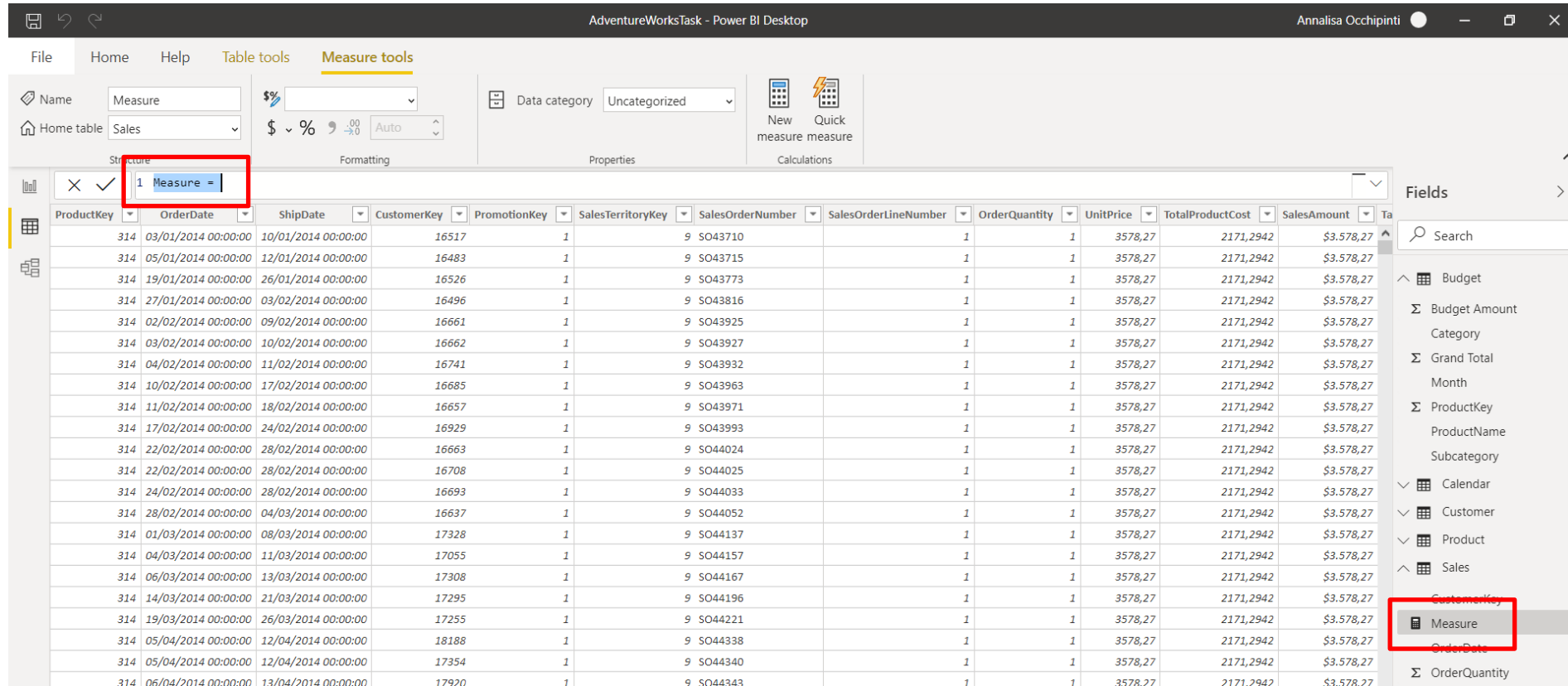
Sometimes you don't need to add a new calculated column, but only a measure (just a number, rather than a full column of numbers). These "single numbers" are called measures.

We are going to add a measure that works out the total of the two columns selected before (the **SalesAmount** and **TaxAmt** columns). Remember that a measure is not "part of the table", so it will not show in the table.

To add a new Measure, click on the Home ribbon and select "New Measure" (Modelling ribbon and "New Measure" for the old versions of Power BI).



Once you click on “New Measure” you will see the word “Measure” will appear in the Formula bar. This measure is also added to the Sales table as shown on the right of the screenshot below.



The screenshot shows the Power BI Desktop interface with the 'AdventureWorksTask' dataset. The 'Measure tools' ribbon is active, and the 'Measure' name is highlighted in the ribbon. The 'Fields' pane on the right shows the 'Sales' table with various fields. The 'Measure' field is highlighted in the Fields pane.

ProductKey	OrderDate	ShipDate	CustomerKey	PromotionKey	SalesTerritoryKey	SalesOrderNumber	SalesOrderLineNumber	OrderQuantity	UnitPrice	TotalProductCost	SalesAmount
314	03/01/2014 00:00:00	10/01/2014 00:00:00	16517	1	9	SO43710	1	1	3578,27	2171,2942	\$3.578,27
314	05/01/2014 00:00:00	12/01/2014 00:00:00	16483	1	9	SO43715	1	1	3578,27	2171,2942	\$3.578,27
314	19/01/2014 00:00:00	26/01/2014 00:00:00	16526	1	9	SO43773	1	1	3578,27	2171,2942	\$3.578,27
314	27/01/2014 00:00:00	03/02/2014 00:00:00	16496	1	9	SO43816	1	1	3578,27	2171,2942	\$3.578,27
314	02/02/2014 00:00:00	09/02/2014 00:00:00	16661	1	9	SO43925	1	1	3578,27	2171,2942	\$3.578,27
314	03/02/2014 00:00:00	10/02/2014 00:00:00	16662	1	9	SO43927	1	1	3578,27	2171,2942	\$3.578,27
314	04/02/2014 00:00:00	11/02/2014 00:00:00	16741	1	9	SO43932	1	1	3578,27	2171,2942	\$3.578,27
314	10/02/2014 00:00:00	17/02/2014 00:00:00	16685	1	9	SO43963	1	1	3578,27	2171,2942	\$3.578,27
314	11/02/2014 00:00:00	18/02/2014 00:00:00	16657	1	9	SO43971	1	1	3578,27	2171,2942	\$3.578,27
314	17/02/2014 00:00:00	24/02/2014 00:00:00	16929	1	9	SO43993	1	1	3578,27	2171,2942	\$3.578,27
314	22/02/2014 00:00:00	28/02/2014 00:00:00	16663	1	9	SO44024	1	1	3578,27	2171,2942	\$3.578,27
314	22/02/2014 00:00:00	28/02/2014 00:00:00	16708	1	9	SO44025	1	1	3578,27	2171,2942	\$3.578,27
314	24/02/2014 00:00:00	28/02/2014 00:00:00	16693	1	9	SO44033	1	1	3578,27	2171,2942	\$3.578,27
314	28/02/2014 00:00:00	04/03/2014 00:00:00	16637	1	9	SO44052	1	1	3578,27	2171,2942	\$3.578,27
314	01/03/2014 00:00:00	08/03/2014 00:00:00	17328	1	9	SO44137	1	1	3578,27	2171,2942	\$3.578,27
314	04/03/2014 00:00:00	11/03/2014 00:00:00	17055	1	9	SO44157	1	1	3578,27	2171,2942	\$3.578,27
314	06/03/2014 00:00:00	13/03/2014 00:00:00	17308	1	9	SO44167	1	1	3578,27	2171,2942	\$3.578,27
314	14/03/2014 00:00:00	21/03/2014 00:00:00	17295	1	9	SO44196	1	1	3578,27	2171,2942	\$3.578,27
314	19/03/2014 00:00:00	26/03/2014 00:00:00	17255	1	9	SO44221	1	1	3578,27	2171,2942	\$3.578,27
314	05/04/2014 00:00:00	12/04/2014 00:00:00	18188	1	9	SO44338	1	1	3578,27	2171,2942	\$3.578,27
314	05/04/2014 00:00:00	12/04/2014 00:00:00	17354	1	9	SO44340	1	1	3578,27	2171,2942	\$3.578,27
314	06/04/2014 00:00:00	13/04/2014 00:00:00	17920	1	9	SO44343	1	1	3578,27	2171,2942	\$3.578,27

If you use the same formula that we used before, this will give you an error. This is because measures are not row-specific, so you need to specify what you want to do with those specific columns. To solve this problem, we use an aggregation function (SUM in this case) to work out the total of each of the two columns and then we will add the two numbers together.

We use the formula:

`SUM(Sales[SalesAmount])` to work out the total of the Sales Amount column,

the formula:

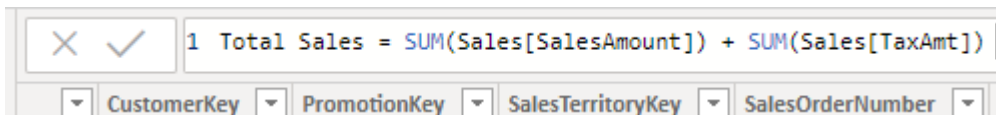
`SUM(Sales[TaxAmt])` to work out the total of the TaxAmt column and

then we add them together with the final formula:

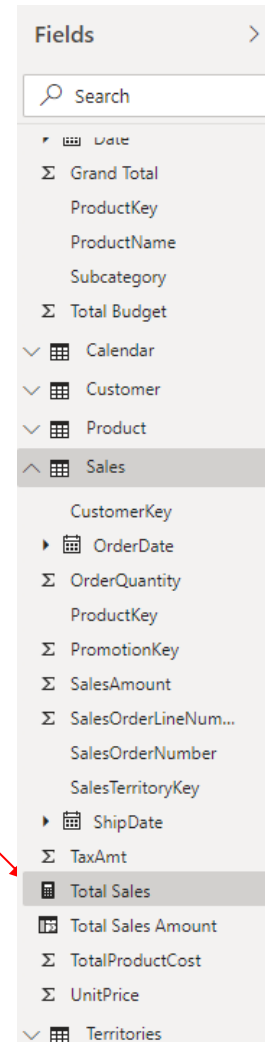
`SUM(Sales[SalesAmount]) + SUM(Sales[TaxAmt])`

Let's do this in Power BI. Type the following formula in the formula bar and press enter.

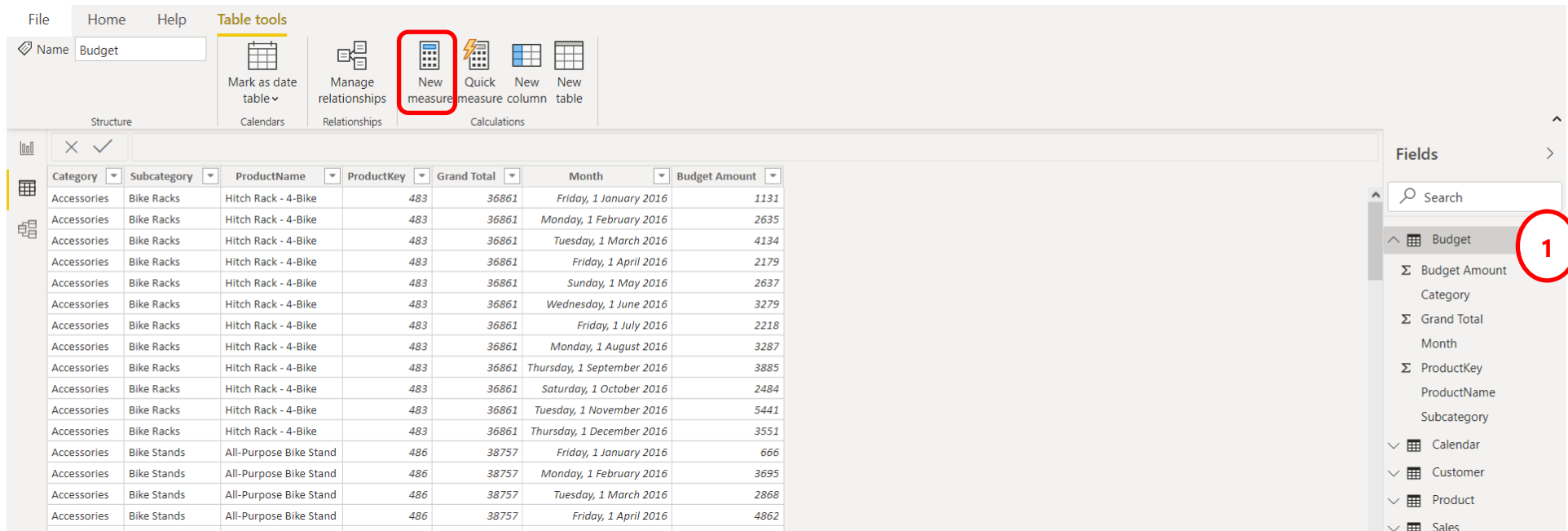
Total Sales = SUM(Sales[SalesAmount]) + SUM(Sales[TaxAmt])



Once you've typed in the formula and pressed enter, you will not see any changes in the table, but a new measure has appeared in the right pane.



The next step is to define a Budget measure to work out the Total of the “Total Budget” column in the Budget table. To do this, we need to create a new measure. Select the **Budget** table on the right pane (see 1 below), and click on New Measure in the Table tools ribbon (if you are working with the old version of Power BI, you can find this tool in the Modelling ribbon)



The screenshot shows the Power BI interface. The 'Table tools' ribbon is active, and the 'New measure' button is highlighted with a red box. The 'Fields' pane on the right shows the 'Budget' table selected, also highlighted with a red circle and the number '1'. The main view displays a table with columns: Category, Subcategory, ProductName, ProductKey, Grand Total, Month, and Budget Amount. The table contains data for various bike accessories and stands across different months in 2016.

Category	Subcategory	ProductName	ProductKey	Grand Total	Month	Budget Amount
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Friday, 1 January 2016	1131
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Monday, 1 February 2016	2635
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Tuesday, 1 March 2016	4134
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Friday, 1 April 2016	2179
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Sunday, 1 May 2016	2637
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Wednesday, 1 June 2016	3279
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Friday, 1 July 2016	2218
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Monday, 1 August 2016	3287
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Thursday, 1 September 2016	3885
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Saturday, 1 October 2016	2484
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Tuesday, 1 November 2016	5441
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Thursday, 1 December 2016	3551
Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	Friday, 1 January 2016	666
Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	Monday, 1 February 2016	3695
Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	Tuesday, 1 March 2016	2868
Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	Friday, 1 April 2016	4862

We are adding a new measure to work out the total of the Budget Amount column. Type the formula

Budget = SUM(Budget[Budget Amount])

in the formula bar and press enter (see below). A new measure has been added to the Budget table (on the right pane).

1 Budget = SUM(Budget[Budget Amount])

Category	Subcategory	ProductName	ProductKey	Grand Total	Month	Budget Amount
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Friday, 1 January 2016	1131
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Monday, 1 February 2016	2635
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Tuesday, 1 March 2016	4134
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Friday, 1 April 2016	2179
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Sunday, 1 May 2016	2637
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Wednesday, 1 June 2016	3279
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Friday, 1 July 2016	2218
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Monday, 1 August 2016	3287
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Thursday, 1 September 2016	3885
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Saturday, 1 October 2016	2484
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Tuesday, 1 November 2016	5441
Accessories	Bike Racks	Hitch Rack - 4-Bike	483	36861	Thursday, 1 December 2016	3551
Accessories	Bike Stands	All-Purpose Bike Stand	486	38757	Friday, 1 January 2016	666

Fields

Search

- Budget
- Budget
- Budget Amount
- Category
- Grand Total
- Month
- ProductKey
- ProductName
- Subcategory

Before moving to the next section, create a new measure in the **Sales** table to work out the total of the SalesAmount column. As we did before, select the **Sales** table, click on New Measure from the Table Tools tab and type the formula below in the formula bar.

Sales = SUM(Sales[SalesAmount])

This formula creates a new measure (a new value) that stores the total of the Sales Amount column. You should see this measure in the right pane, under the Sales table.

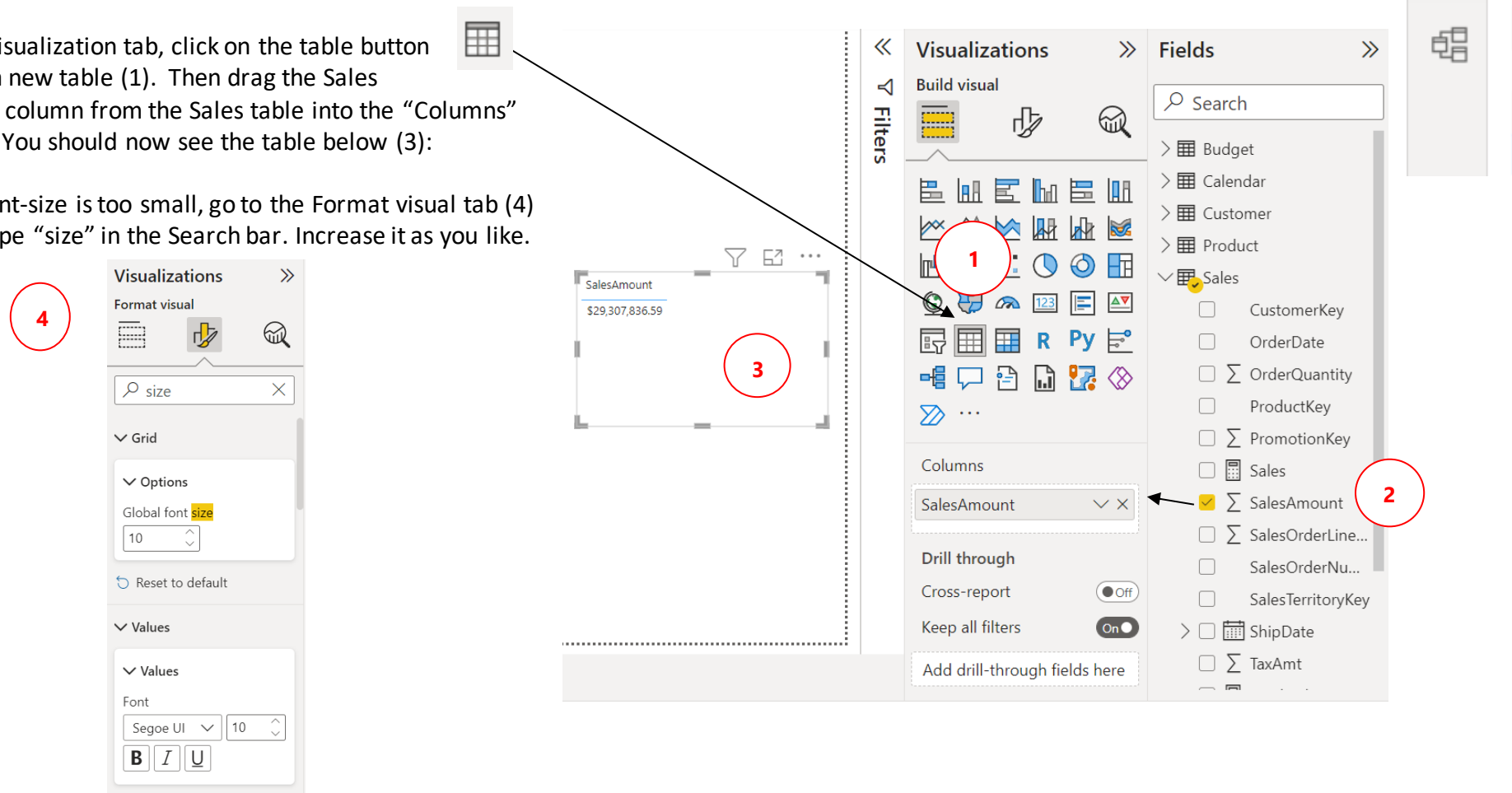
Product
Sales
CustomerKey
OrderDate
OrderQuantity
ProductKey
PromotionKey
Sales
SalesAmount
SalesOrderLineNu...

Step 3: Adding the Visual Tables

The next step is to use the Budget Measure to investigate the data and perform more Data Analytics tasks. Click on the report tab on the left hand-side of the main Power BI window. We want to add a table to have a better look at the data using the measures and calculated columns added so far.

In the Visualization tab, click on the table button (1). Then drag the Sales amount column from the Sales table into the “Columns” box (2). You should now see the table below (3):

If the font-size is too small, go to the Format visual tab (4) and type “size” in the Search bar. Increase it as you like.



The screenshot illustrates the steps to add a table visual in Power BI:

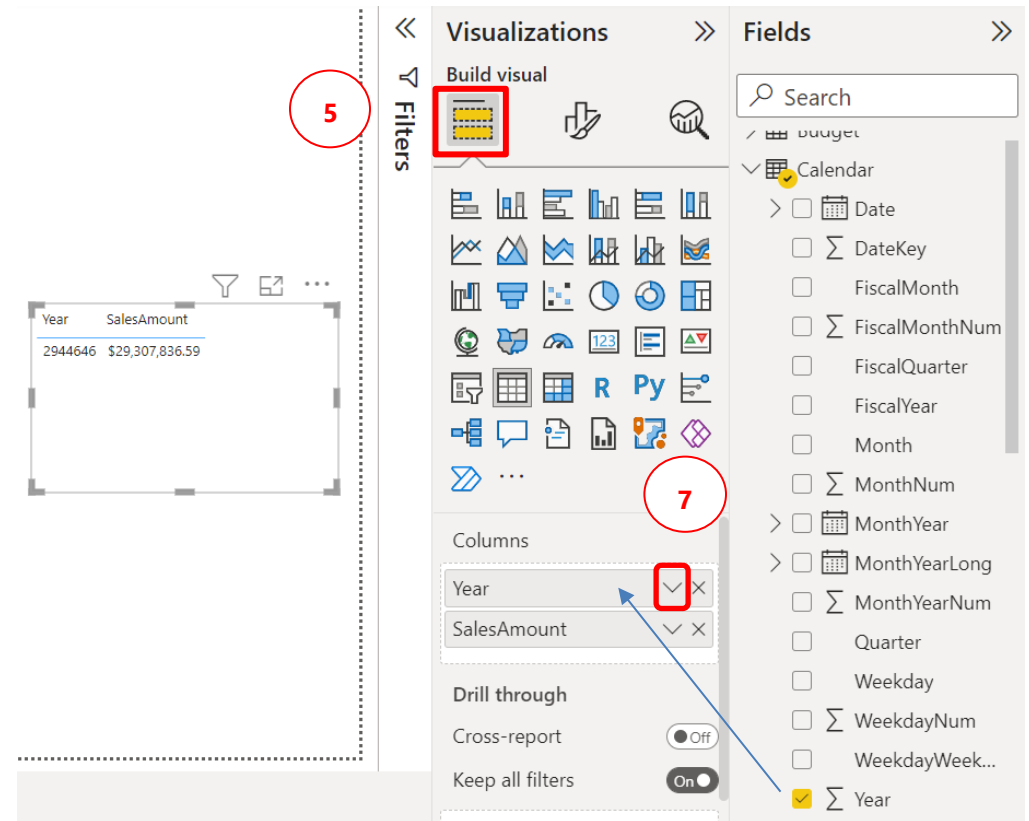
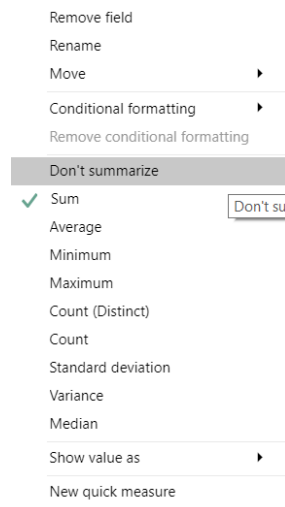
- Step 1:** In the Visualizations pane, click on the table icon (represented by a grid).
- Step 2:** Drag the SalesAmount column from the Sales table in the Fields pane into the Columns box.
- Step 3:** The resulting table visual is displayed, showing the SalesAmount value of \$29,307,836.59.
- Step 4:** In the Format visual pane, search for "size" and increase the font size to make the text larger.

The next step is to use the Calendar table (with the relationships we created before) to slice the Sales Amount. Let's start by slicing per year. Go back to the Build Visual tab (5). Drag the "Year" column from the Calendar table on the right to the Columns field (6). Drop the Year column above the SalesAmount, in this way the table will show the Year column first and the SalesAmount column second.

The number displayed in the table is actually the "sum of Calendar year", which is not what we need. We can change this by clicking on the drop down menu next to the Year tab (in the Columns box – (7)) and select "Don't summarize".

Your new table should now look like the one below

Year	SalesAmount
2014	\$7,072,084.24
2015	\$5,762,134.30
2016	\$16,473,618.05
Total	\$29,307,836.59

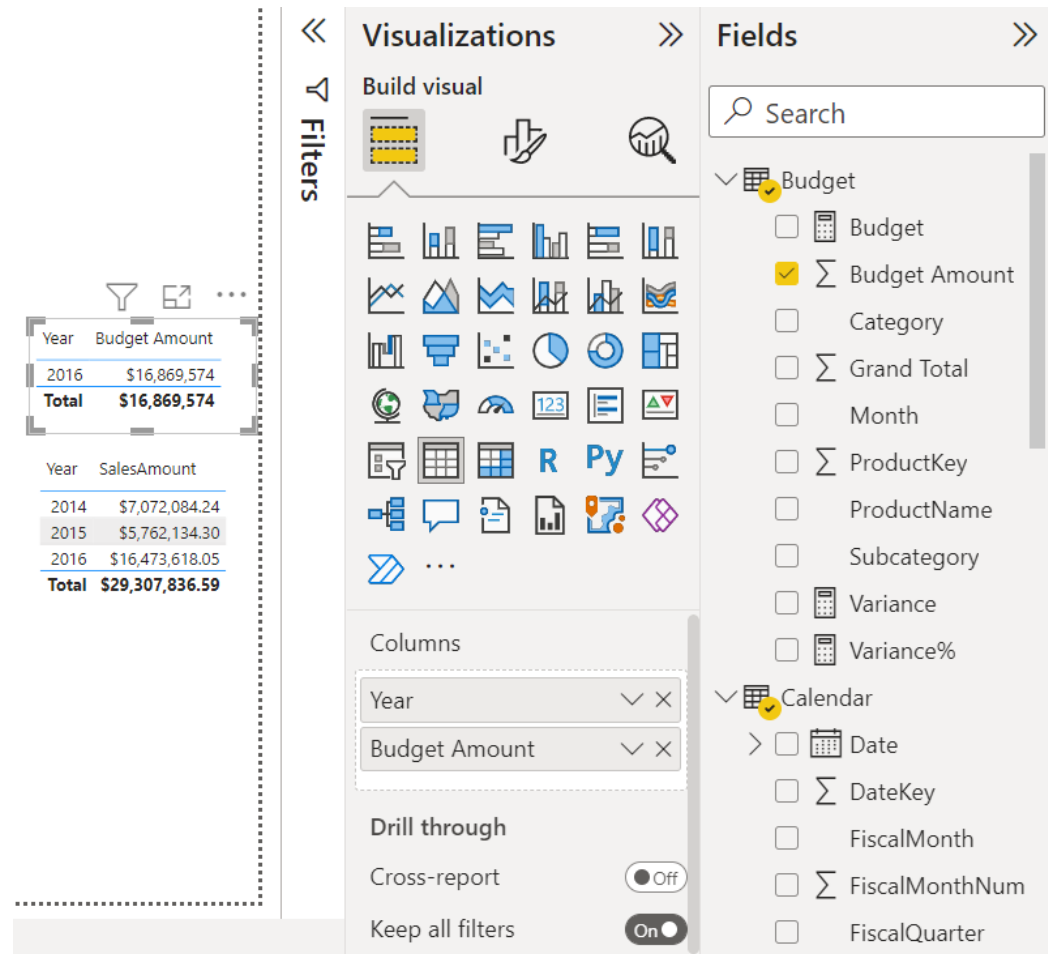


Now, let's repeat the steps above to create a Budget table. Make sure the previous table is not selected before doing the steps below. You can deselect the table by clicking anywhere in the blank canvas of the Report view.

- 1) Select the Table symbol from the Visualization pane 
- 2) Drag the **Budget Amount** column from the **Budget** table and drop it in the Columns field.
- 3) Drag the **Year** column from the **Calendar** table and drop it in the Columns field, above the Budget Amount value.
- 4) Select the "Don't summarize" option.
- 5) You should now have the two tables on the right.

The Budget Amount table shows the total amount of Budget just for 2016 because that is the only year available in that table. Since we want to compare the two tables, we need to focus only on the Sales related to 2016 in the SalesAmount table.

To do this, we need add some filters.



The screenshot shows the report builder interface. On the left, two tables are displayed:

Year	Budget Amount
2016	\$16,869,574
Total	\$16,869,574

Year	SalesAmount
2014	\$7,072,084.24
2015	\$5,762,134.30
2016	\$16,473,618.05
Total	\$29,307,836.59

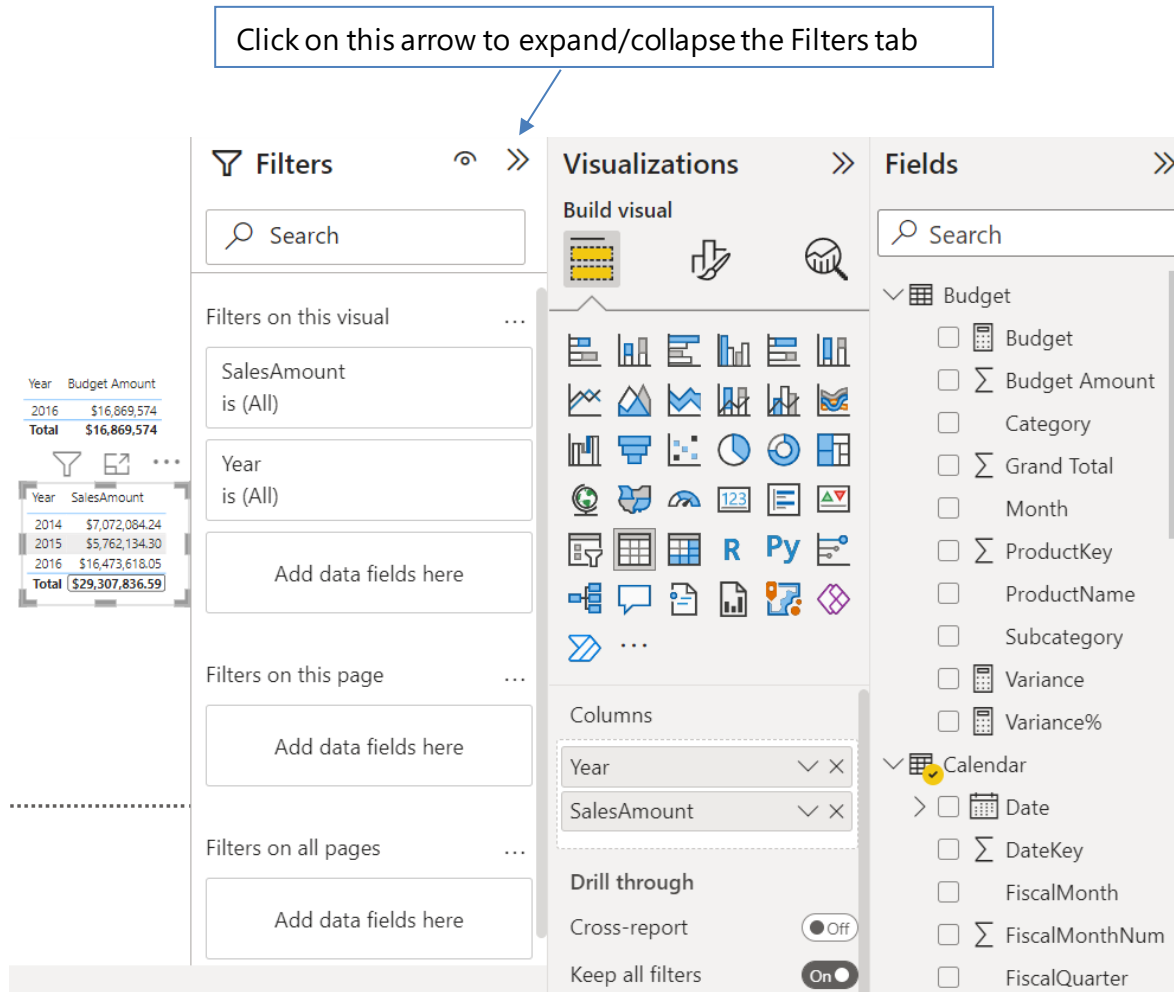
On the right, the 'Visualizations' pane shows the 'Build visual' section with various chart icons. Below it, the 'Columns' field contains 'Year' and 'Budget Amount'. The 'Filters' section has 'Cross-report' set to 'Off' and 'Keep all filters' set to 'On'.

The 'Fields' pane on the far right shows the available fields for the 'Budget' and 'Calendar' tables. The 'Budget' table fields are: Budget, Budget Amount (checked), Category, Grand Total, Month, ProductKey, ProductName, Subcategory, Variance, and Variance%. The 'Calendar' table fields are: Date (checked), DateKey, FiscalMonth, FiscalMonthNum, and FiscalQuarter.

Step 4: Adding Filters

To do add the 2016 filter, click on the SalesAmount table and expand the Filters tab.

Click on this arrow to expand/collapse the Filters tab



Filters

Search

Filters on this visual ...

SalesAmount is (All)

Year is (All)

Add data fields here

Filters on this page ...

Add data fields here

Filters on all pages ...

Add data fields here

Visualizations

Build visual

Columns

Year

SalesAmount

Drill through

Cross-report Off

Keep all filters On

Fields

Search

Budget

- ☐ Budget
- ☐ Budget Amount
- ☐ Category
- ☐ Grand Total
- ☐ Month
- ☐ ProductKey
- ☐ ProductName
- ☐ Subcategory
- ☐ Variance
- ☐ Variance%

Calendar

- ☐ Date
- ☐ DateKey
- ☐ FiscalMonth
- ☐ FiscalMonthNum
- ☐ FiscalQuarter

Year	Budget Amount
2016	\$16,869,574
Total	\$16,869,574

Year	SalesAmount
2014	\$7,072,084.24
2015	\$5,762,134.30
2016	\$16,473,618.05
Total	\$29,307,836.59

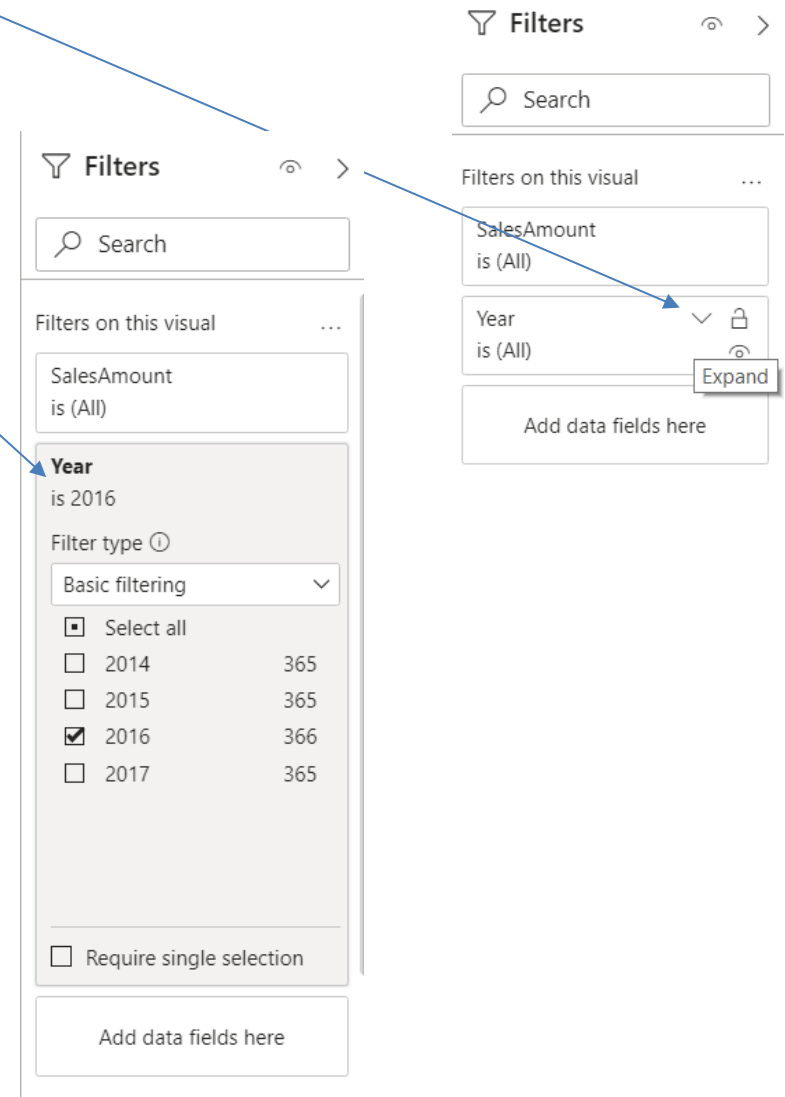
1) Use the drop down menu next to Year to open the Filtering options. The drop-down arrow will appear when you go with your mouse over to the Year section.

2) In the filtering options, change the Filter type as Basic Filtering, then select 2016.

3) Your table is now showing only the Sales in 2016.

Year	Budget Amount
2016	\$16,869,574
Total	\$16,869,574

Year	SalesAmount
2016	\$16,473,618.05
Total	\$16,473,618.05



The screenshot shows the Power BI Filters pane with two filter cards. The 'SalesAmount' card is set to 'is (All)'. The 'Year' card is expanded, showing 'is 2016' and 'Filter type' set to 'Basic filtering'. Under 'Basic filtering', the 'Select all' checkbox is checked, and the year 2016 is selected with a checkmark. The 'Require single selection' checkbox is unchecked. An 'Expand' button is visible next to the Year card.

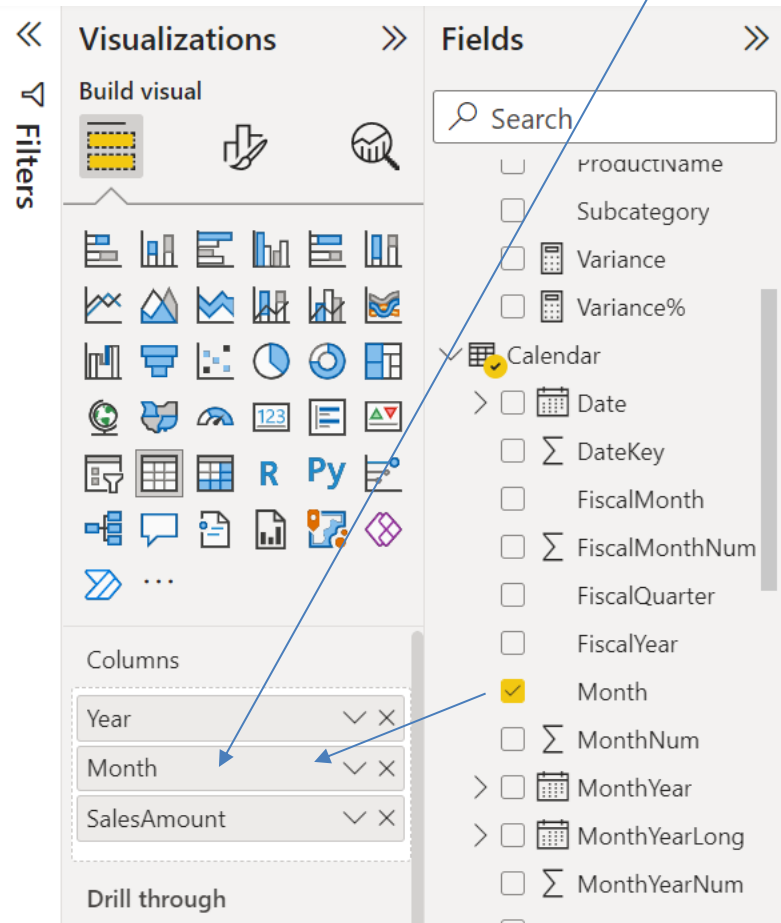
4) The next step is to add the Months, rather than just the Year.

Select the Budget Amount table. In the visualization tab, drag and drop the Months column from the Calendar table into the Columns tab (in between the Year and SalesAmount values).

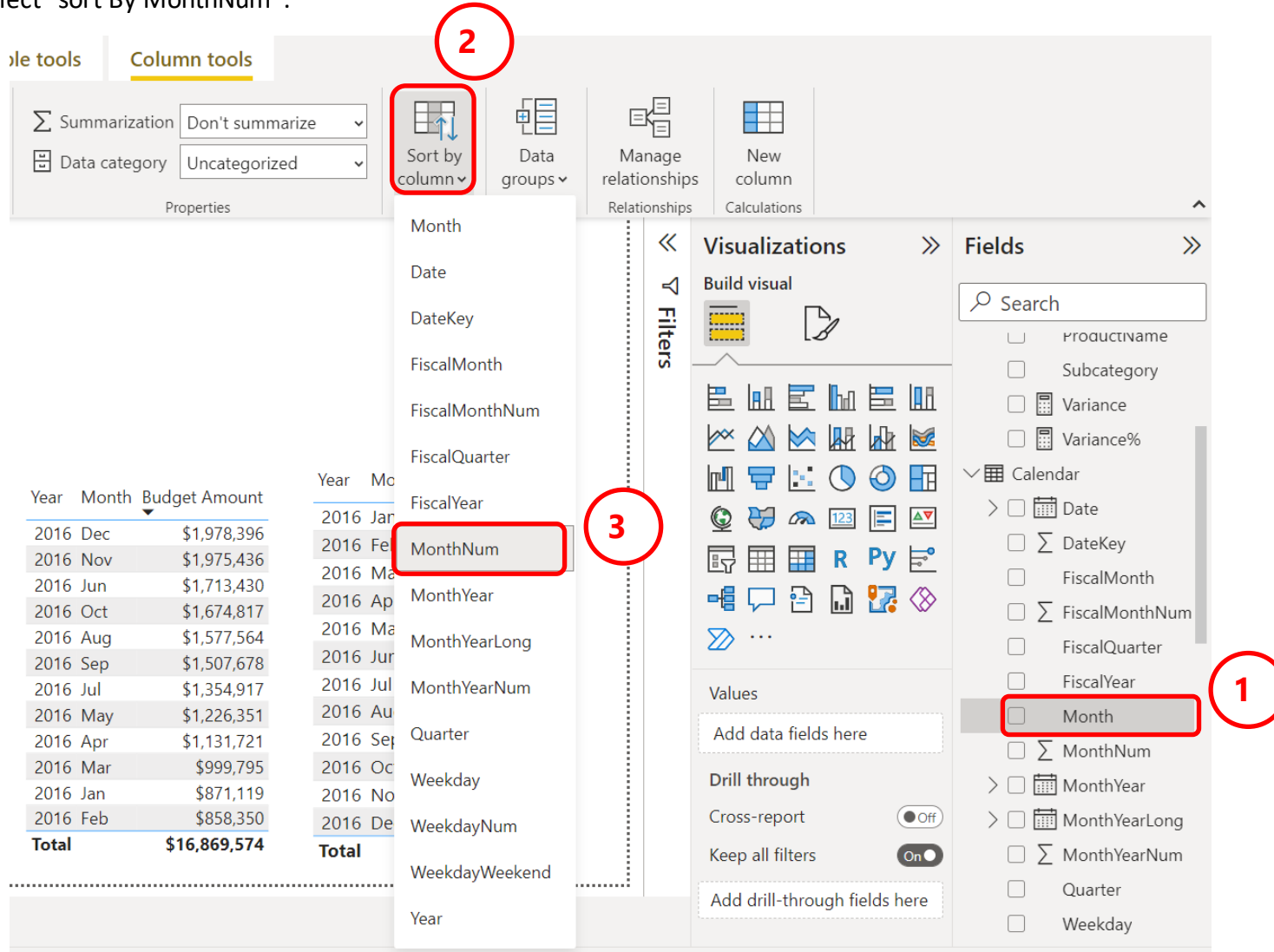
5) Repeat step 4) for the Sales Amount table. The resulting tables are shown below.

Year	Month	Budget Amount
2016	Dec	\$1,978,396
2016	Nov	\$1,975,436
2016	Jun	\$1,713,430
2016	Oct	\$1,674,817
2016	Aug	\$1,577,564
2016	Sep	\$1,507,678
2016	Jul	\$1,354,917
2016	May	\$1,226,351
2016	Apr	\$1,131,721
2016	Mar	\$999,795
2016	Jan	\$871,119
2016	Feb	\$858,350
Total		\$16,869,574

Year	Month	SalesAmount
2016	Jan	\$886,668.84
2016	Feb	\$847,413.51
2016	Mar	\$1,010,258.13
2016	Apr	\$1,080,449.58
2016	May	\$1,196,981.11
2016	Jun	\$1,731,787.77
2016	Jul	\$1,340,244.95
2016	Aug	\$1,462,479.83
2016	Sep	\$1,480,905.18
2016	Oct	\$1,608,750.53
2016	Nov	\$1,878,317.51
2016	Dec	\$1,949,361.11
Total		\$16,473,618.05



- 6) To sort the Months in the table (starting from January), click on the Month column in the Fields panel (1). Then in the Column Tools tab, select Sort by Column (2) and click on "MonthNum" (3). If you are working with an old version of Power BI, go to Sort by Column in the Modelling ribbon and select "sort By MonthNum".



The screenshot shows the Power BI interface with the following elements:

- Column tools tab:** The 'Sort by column' button is highlighted with a red circle labeled '2'.
- Fields panel:** The 'Month' field is highlighted with a red circle labeled '1'.
- Sort dropdown menu:** The 'MonthNum' option is highlighted with a red circle labeled '3'.

The table below shows the data for 2016, sorted by Month (starting from January):

Year	Month	Budget Amount
2016	Dec	\$1,978,396
2016	Nov	\$1,975,436
2016	Jun	\$1,713,430
2016	Oct	\$1,674,817
2016	Aug	\$1,577,564
2016	Sep	\$1,507,678
2016	Jul	\$1,354,917
2016	May	\$1,226,351
2016	Apr	\$1,131,721
2016	Mar	\$999,795
2016	Jan	\$871,119
2016	Feb	\$858,350
Total		\$16,869,574

The final tables are shown below.

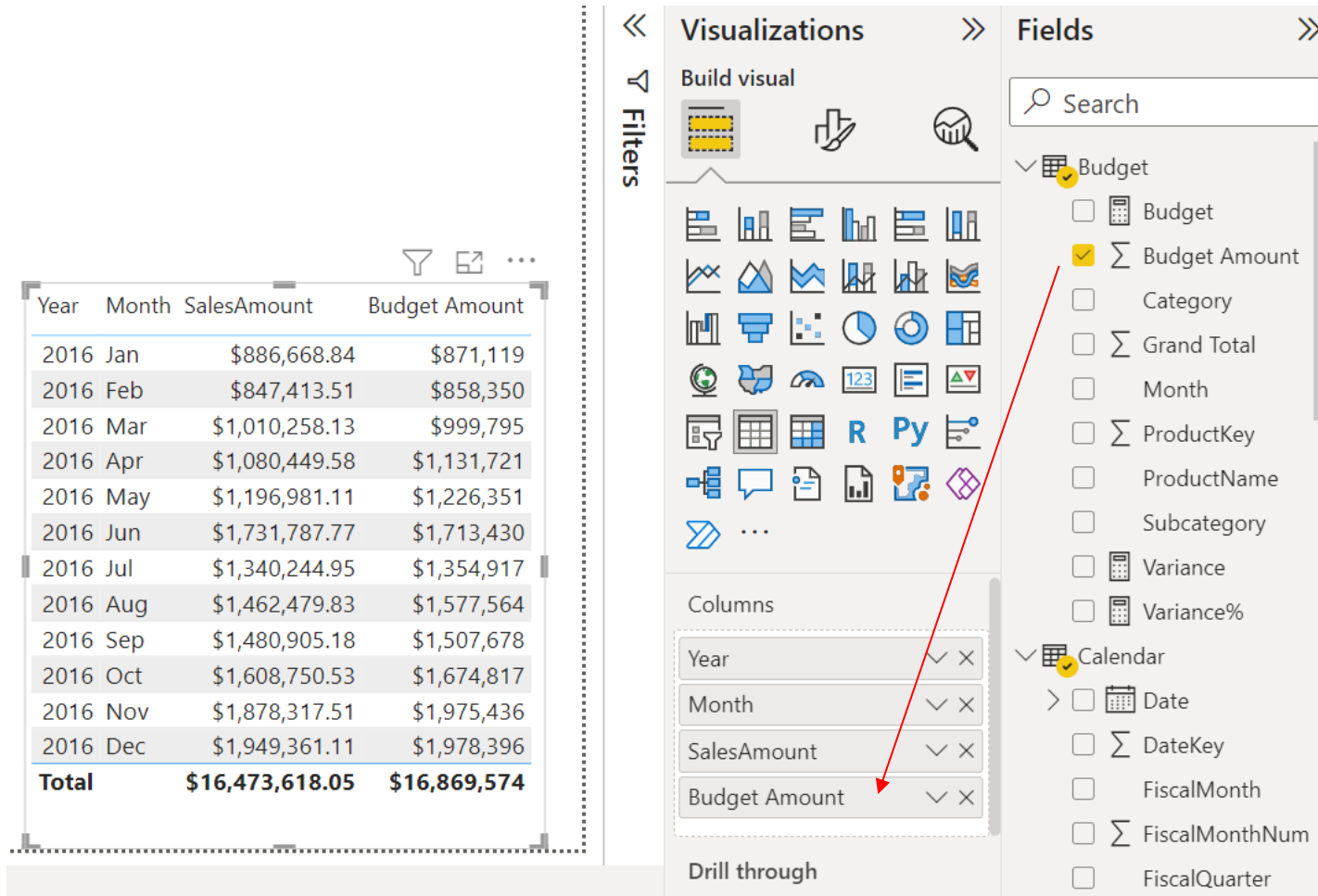
Year	Month	Budget Amount
2016	Jan	\$871,119
2016	Feb	\$858,350
2016	Mar	\$999,795
2016	Apr	\$1,131,721
2016	May	\$1,226,351
2016	Jun	\$1,713,430
2016	Jul	\$1,354,917
2016	Aug	\$1,577,564
2016	Sep	\$1,507,678
2016	Oct	\$1,674,817
2016	Nov	\$1,975,436
2016	Dec	\$1,978,396
Total		\$16,869,574

Year	Month	SalesAmount
2016	Jan	\$886,668.84
2016	Feb	\$847,413.51
2016	Mar	\$1,010,258.13
2016	Apr	\$1,080,449.58
2016	May	\$1,196,981.11
2016	Jun	\$1,731,787.77
2016	Jul	\$1,340,244.95
2016	Aug	\$1,462,479.83
2016	Sep	\$1,480,905.18
2016	Oct	\$1,608,750.53
2016	Nov	\$1,878,317.51
2016	Dec	\$1,949,361.11
Total		\$16,473,618.05

They now look much better. However, if you want to compare the Sales Amount and the Budget Amount, it would be nice to have them side by side in the same table. Let's see how to do this in the next step. After we have merged both tables into one, we will create a Variance column using DAX to further investigate the difference between the BudgetAmount and SalesAmount.

Step 5: Adding a Variance Table using DAX

To merge the two tables, select the SalesAmount table (click on the table) and add the BudgetAmount column in the Columns box. The final table is shown below. You can now compare Sales and Budget month by month in the same table. Please note that all this has been possible because of the relationships between tables that we created before. Both the Sales and Budget tables are connected to the Date table, so they can be connected to each other.

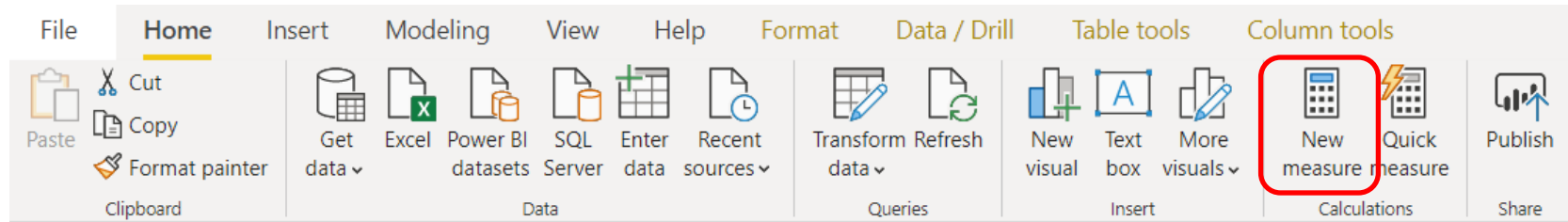


The screenshot displays a Power BI Desktop interface. On the left, a table visual shows monthly sales and budget data for 2016. The table has four columns: Year, Month, SalesAmount, and Budget Amount. The data is organized by month from January to December, with a total row at the bottom. The SalesAmount and Budget Amount columns are formatted with currency symbols and commas. On the right, the Fields pane is open, showing the Budget and Calendar tables. The Budget table is expanded, and the Budget Amount column is selected. The Calendar table is also expanded, and the Date column is selected. A red arrow points from the Budget Amount column in the Fields pane to the Budget Amount column in the table visual. The Columns box in the Visualizations pane shows the fields Year, Month, SalesAmount, and Budget Amount.

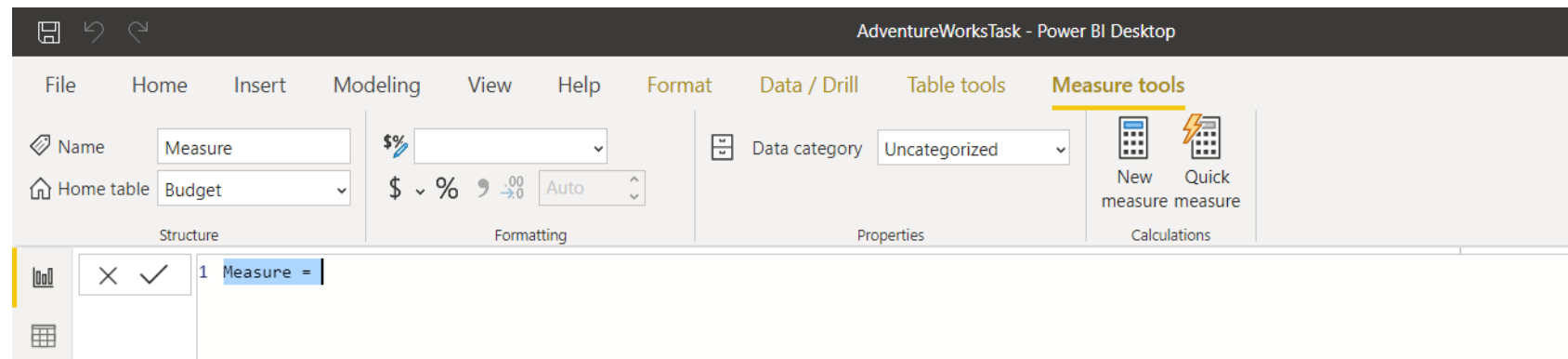
Year	Month	SalesAmount	Budget Amount
2016	Jan	\$886,668.84	\$871,119
2016	Feb	\$847,413.51	\$858,350
2016	Mar	\$1,010,258.13	\$999,795
2016	Apr	\$1,080,449.58	\$1,131,721
2016	May	\$1,196,981.11	\$1,226,351
2016	Jun	\$1,731,787.77	\$1,713,430
2016	Jul	\$1,340,244.95	\$1,354,917
2016	Aug	\$1,462,479.83	\$1,577,564
2016	Sep	\$1,480,905.18	\$1,507,678
2016	Oct	\$1,608,750.53	\$1,674,817
2016	Nov	\$1,878,317.51	\$1,975,436
2016	Dec	\$1,949,361.11	\$1,978,396
Total		\$16,473,618.05	\$16,869,574

We are now ready to add a variance column. We are going to create a “Variance” measure and then we will add this measure to the table. Follow the steps below to create and add the new measure.

- 1) Select the Budget table from the right pane (click on it).
- 2) From the Home ribbon, select “New Measure

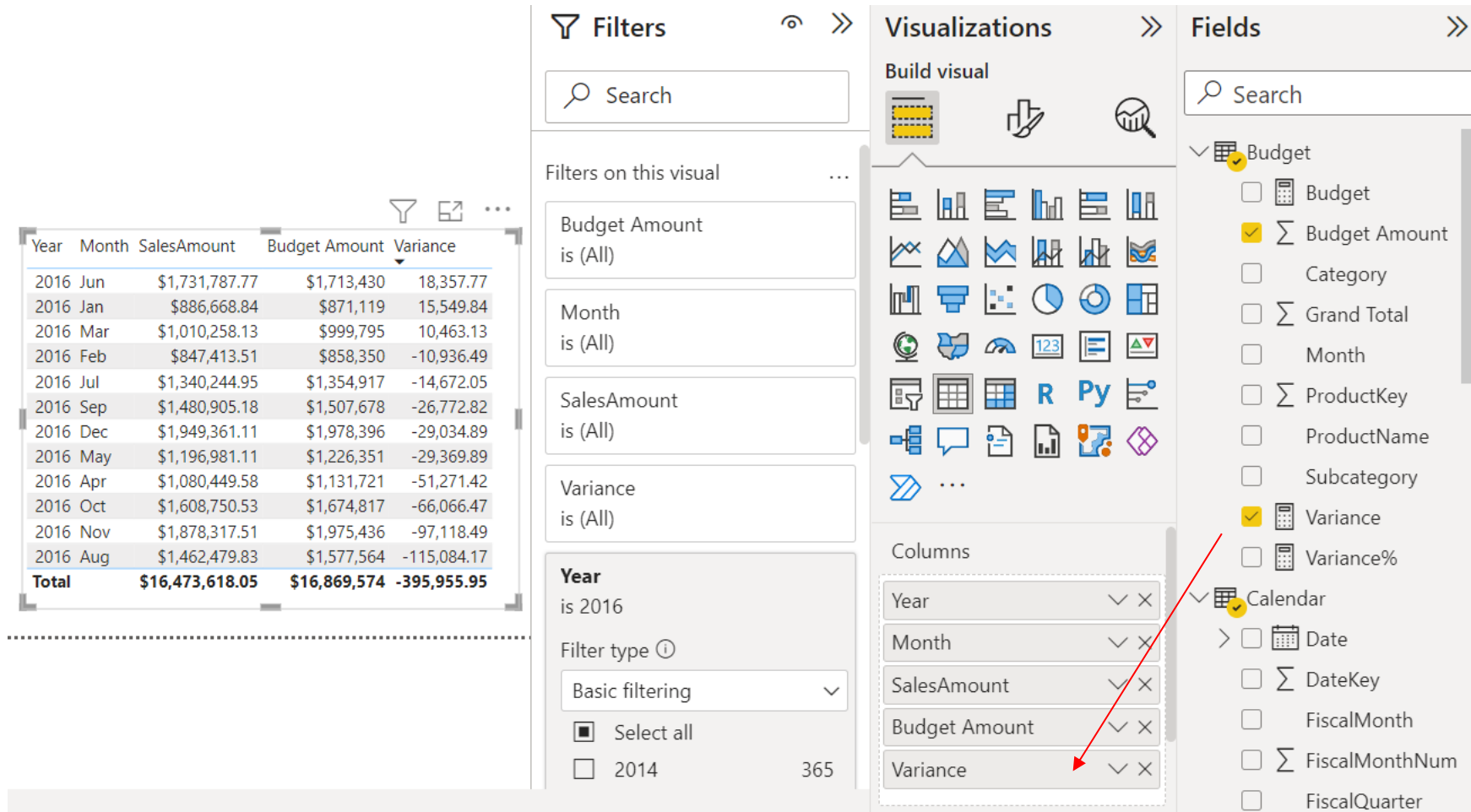


- 3) In the formula bar at the top, delete “Measure = ” and replace it with the formula in Step 4)



- 4) `Variance = [Sales] - [Budget]` This formula uses the Sales and Budget measures we created before. These two measures are necessary if you want to compare two columns (Budget Amount and Sales Amount) that are from two different tables not directly connected to each other (the Budget and Sales tables). Enter the formula and press Enter. Click on any of the tables on the right pane (Fields), if you can't easily go back to the Report view.

5) To add this new measure in your table, drag the Variance column from the Fields to the Columns box (see below).



The screenshot displays the Power BI Desktop interface with a table view on the left and the Fields pane on the right. The table shows monthly sales and budget data for 2016, including a 'Variance' column. The Fields pane shows the 'Budget' table with 'Budget Amount' and 'Variance' measures. A red arrow points from the 'Variance' measure in the Fields pane to the 'Columns' box in the Visualizations pane, indicating the action to add it to the table.

Year	Month	SalesAmount	Budget Amount	Variance
2016	Jun	\$1,731,787.77	\$1,713,430	18,357.77
2016	Jan	\$886,668.84	\$871,119	15,549.84
2016	Mar	\$1,010,258.13	\$999,795	10,463.13
2016	Feb	\$847,413.51	\$858,350	-10,936.49
2016	Jul	\$1,340,244.95	\$1,354,917	-14,672.05
2016	Sep	\$1,480,905.18	\$1,507,678	-26,772.82
2016	Dec	\$1,949,361.11	\$1,978,396	-29,034.89
2016	May	\$1,196,981.11	\$1,226,351	-29,369.89
2016	Apr	\$1,080,449.58	\$1,131,721	-51,271.42
2016	Oct	\$1,608,750.53	\$1,674,817	-66,066.47
2016	Nov	\$1,878,317.51	\$1,975,436	-97,118.49
2016	Aug	\$1,462,479.83	\$1,577,564	-115,084.17
Total		\$16,473,618.05	\$16,869,574	-395,955.95

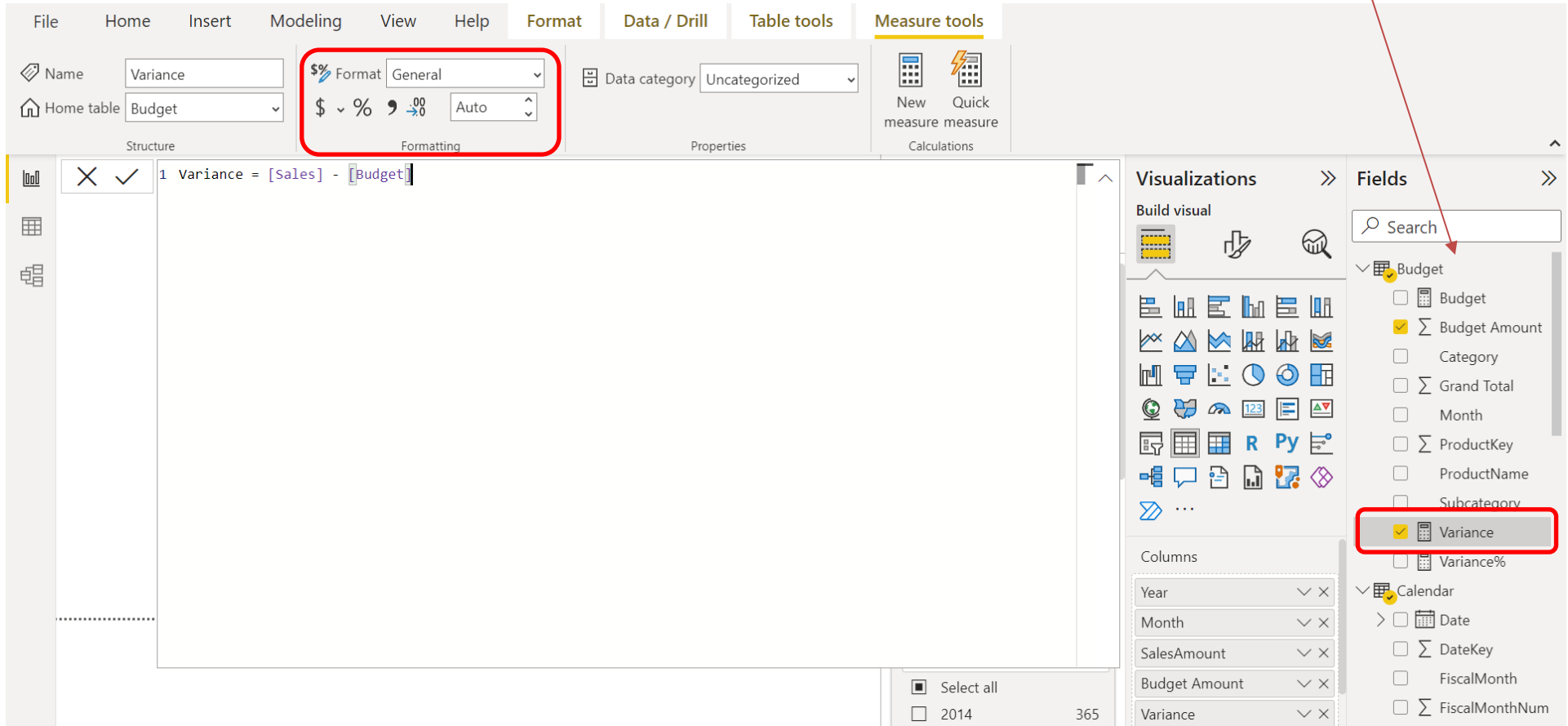
Fields Pane:

- Budget**
 - ☐ Budget
 - ☒ Σ Budget Amount
 - ☐ Category
 - ☐ Σ Grand Total
 - ☐ Month
 - ☐ Σ ProductKey
 - ☐ ProductName
 - ☐ Subcategory
 - ☒ Σ Variance
 - ☐ Variance%
- Calendar**
 - ☐ Date
 - ☐ Σ DateKey
 - ☐ FiscalMonth
 - ☐ Σ FiscalMonthNum
 - ☐ FiscalQuarter

Columns:

- Year
- Month
- SalesAmount
- Budget Amount
- Variance

- 6) The new table is now showing the difference between the Sales and Budget amounts month by month. To add the "\$" symbol to the variance column, click on the Variance Measure in the right pane. Under the Measure Tools tab, select the drop-down menu next to the "\$" symbol and click on "\$ English (United States)." To go back to the Report view and see the changes, click on any of the tables on the right.



The screenshot shows the Microsoft Power BI Desktop interface. The top ribbon includes tabs for File, Home, Insert, Modeling, View, Help, Format, Data / Drill, Table tools, and Measure tools. The Measure tools tab is active, showing options for New measure and Quick measure. The main area displays the DAX formula: `1 Variance = [Sales] - [Budget]`. The right pane is divided into Visualizations and Fields. The Fields pane shows a list of fields, with 'Variance' highlighted under the 'Budget' category. A red box highlights the 'Variance' field in the Fields pane. A red arrow points from the text in the instruction block to the 'Variance' field.

- 7) The final table is shown on the right. Note that when the variance is negative, Power BI adds brackets around each value.
- 8) In the next step, we will see how to add a Percentage Variance column, as the Variance value divided by the Budget amount. This will give us a new column to check how the difference between Sales and Budget (Variance) is changing in relation to the Budget amount.

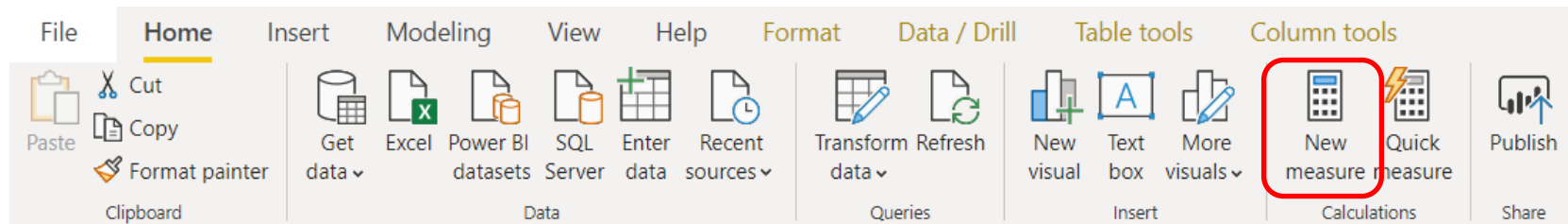
Year	Month	SalesAmount	Budget Amount	Variance
2016	Jan	\$886,668.84	\$871,119	\$15,549.84
2016	Feb	\$847,413.51	\$858,350	(\$10,936.49)
2016	Mar	\$1,010,258.13	\$999,795	\$10,463.13
2016	Apr	\$1,080,449.58	\$1,131,721	(\$51,271.42)
2016	May	\$1,196,981.11	\$1,226,351	(\$29,369.89)
2016	Jun	\$1,731,787.77	\$1,713,430	\$18,357.77
2016	Jul	\$1,340,244.95	\$1,354,917	(\$14,672.05)
2016	Aug	\$1,462,479.83	\$1,577,564	(\$115,084.17)
2016	Sep	\$1,480,905.18	\$1,507,678	(\$26,772.82)
2016	Oct	\$1,608,750.53	\$1,674,817	(\$66,066.47)
2016	Nov	\$1,878,317.51	\$1,975,436	(\$97,118.49)
2016	Dec	\$1,949,361.11	\$1,978,396	(\$29,034.89)
Total		\$16,473,618.05	\$16,869,574	(\$395,955.95)

Step 6: Adding a Percentage Variance Measure

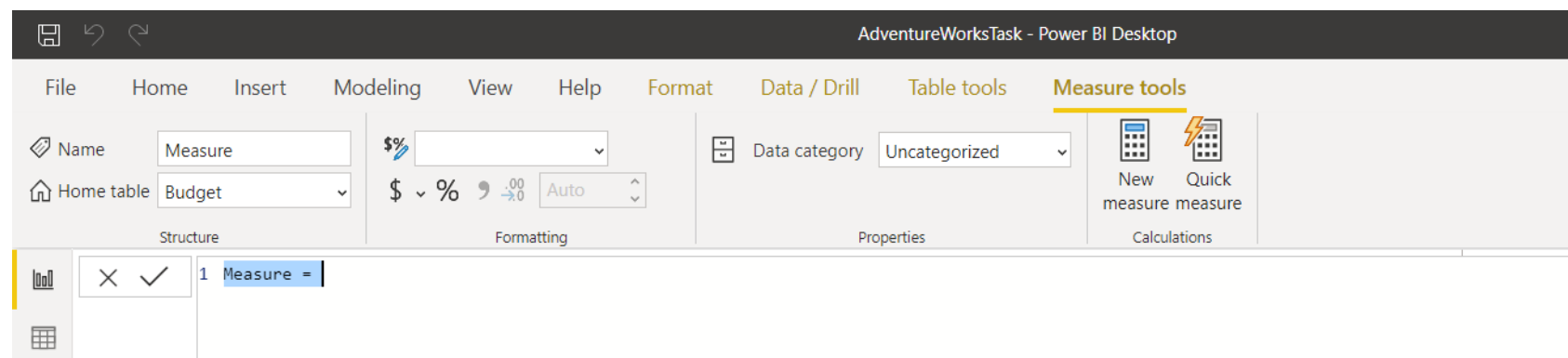
Let's now add the Percentage Variance measure to the check the change between the Sales and Budget over the Budget Amount.

Let's repeat the steps done before to add the new measure.

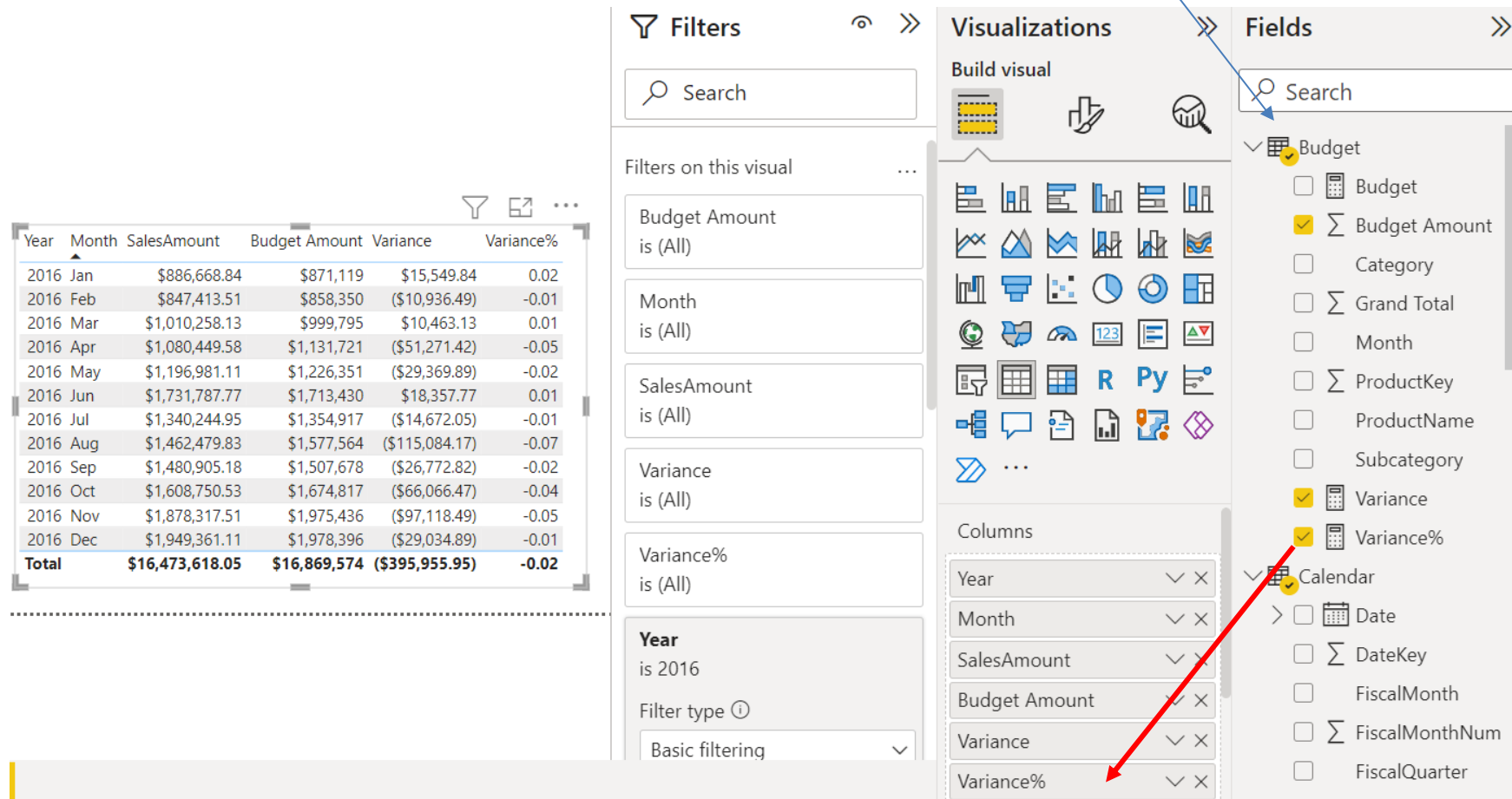
- 1) Select the Budget table from the right pane (click on it).
- 2) From the Home ribbon, select "New Measure"



- 3) In the formula bar at the top, delete "Measure = " and replace it with the formula in Step 4)

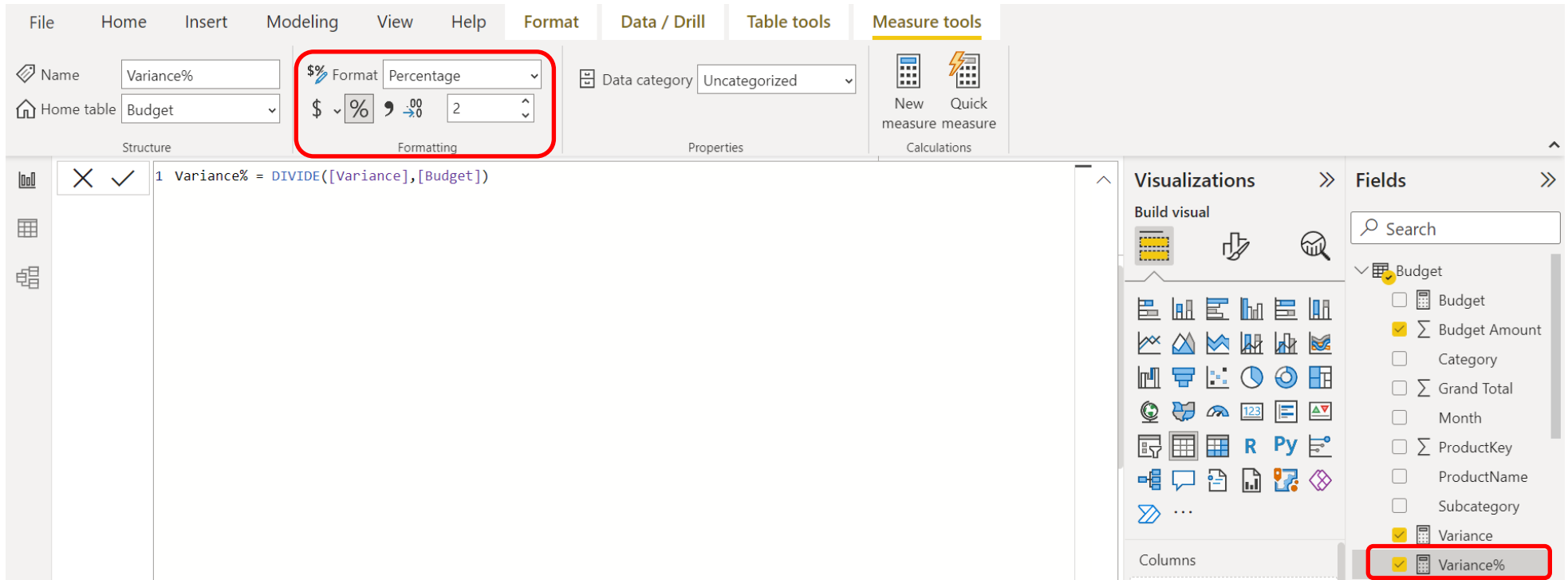


- 4) $\text{Variance\%} = \text{DIVIDE}([\text{Variance}], [\text{Budget}])$ This formula uses the Variance and Budget measures we created before. The DIVIDE DAX function is used to divide Variance by Budget. This will allow to investigate the Variance as a percentage of Budget. Copy this formula in the formula bar and press enter. If you can't close the formula bar, click on any of the tables in the right pane.
- 5) To add this new measure in your table, drag the Variance% column from the Fields to the Columns box. This will add the new **Variance%** column to the table (see below).



Year	Month	SalesAmount	Budget Amount	Variance	Variance%
2016	Jan	\$886,668.84	\$871,119	\$15,549.84	0.02
2016	Feb	\$847,413.51	\$858,350	(\$10,936.49)	-0.01
2016	Mar	\$1,010,258.13	\$999,795	\$10,463.13	0.01
2016	Apr	\$1,080,449.58	\$1,131,721	(\$51,271.42)	-0.05
2016	May	\$1,196,981.11	\$1,226,351	(\$29,369.89)	-0.02
2016	Jun	\$1,731,787.77	\$1,713,430	\$18,357.77	0.01
2016	Jul	\$1,340,244.95	\$1,354,917	(\$14,672.05)	-0.01
2016	Aug	\$1,462,479.83	\$1,577,564	(\$115,084.17)	-0.07
2016	Sep	\$1,480,905.18	\$1,507,678	(\$26,772.82)	-0.02
2016	Oct	\$1,608,750.53	\$1,674,817	(\$66,066.47)	-0.04
2016	Nov	\$1,878,317.51	\$1,975,436	(\$97,118.49)	-0.05
2016	Dec	\$1,949,361.11	\$1,978,396	(\$29,034.89)	-0.01
Total		\$16,473,618.05	\$16,869,574	(\$395,955.95)	-0.02

- 6) The last step is to change the format of the column as Percentages values. To do this, select the Variance% column from the right pane. In the Measure tools tab, click on the “%” symbol in the Formatting section. If your formula bar is covering the whole Report canvas (like mine does below), click on any of the tables on the right side to close the formula bar and go back to Report view. If you are using an old version of Power BI, you can find this under the Modelling tab (Formatting section).



The screenshot shows the Power BI Desktop interface. The top ribbon has tabs: File, Home, Insert, Modeling, View, Help, **Format**, Data / Drill, Table tools, and **Measure tools**. The **Measure tools** tab is active, showing the 'Format' section with a dropdown menu set to 'Percentage'. Below this, there are icons for currency (\$), percentage (%), and a decimal separator (comma/period), along with a decimal places spinner set to 2. The 'Data category' dropdown is set to 'Uncategorized'. The 'Structure' pane on the left shows the formula bar with the measure: `1 Variance% = DIVIDE([Variance],[Budget])`. The 'Visualizations' pane on the right shows various chart types. The 'Fields' pane on the right shows the 'Budget' table expanded, with fields like Budget, Budget Amount, Category, Grand Total, Month, ProductKey, ProductName, Subcategory, Variance, and **Variance%** (which is highlighted with a red box). The 'Columns' section at the bottom of the Fields pane also shows **Variance%** highlighted with a red box.

The final table is shown on the right. The **Variance%** column shows negative values for all the months with a negative Variance (the values in brackets are all negative).

This new column can be used to analyse how the difference between the Sales and the Budget (variance) has changed as a percentage of the Budget amount.

Year	Month	SalesAmount	Budget Amount	Variance	Variance%
2016	Jan	\$886,668.84	\$871,119	\$15,549.84	1.79%
2016	Feb	\$847,413.51	\$858,350	(\$10,936.49)	-1.27%
2016	Mar	\$1,010,258.13	\$999,795	\$10,463.13	1.05%
2016	Apr	\$1,080,449.58	\$1,131,721	(\$51,271.42)	-4.53%
2016	May	\$1,196,981.11	\$1,226,351	(\$29,369.89)	-2.39%
2016	Jun	\$1,731,787.77	\$1,713,430	\$18,357.77	1.07%
2016	Jul	\$1,340,244.95	\$1,354,917	(\$14,672.05)	-1.08%
2016	Aug	\$1,462,479.83	\$1,577,564	(\$115,084.17)	-7.30%
2016	Sep	\$1,480,905.18	\$1,507,678	(\$26,772.82)	-1.78%
2016	Oct	\$1,608,750.53	\$1,674,817	(\$66,066.47)	-3.94%
2016	Nov	\$1,878,317.51	\$1,975,436	(\$97,118.49)	-4.92%
2016	Dec	\$1,949,361.11	\$1,978,396	(\$29,034.89)	-1.47%
Total		\$16,473,618.05	\$16,869,574	(\$395,955.95)	-2.35%

Well done! You have just used the relationships between two tables to “slice-and-dice” them together. But you have also defined new DAX measures to analyse the difference between the two columns. You have also seen how new measures can be defined in Data view, Report view or Power Query Editor but they can be used anywhere else across your model, and this is what makes Power BI so powerful. The next step is to add visuals to create a report, which will be done in the next lesson. Make sure you save this file as “AdventureWorksTask”, you will need this in one of the next lessons when creating reports.